

"WAN Essentials"

- **WAN (Wide Area Network)**
 - A WAN connects LANs across long distances.
 - It uses different transmission methods depending on the needs of the organization.
- **WAN vs. LAN Differences**
 - ! ◦ LANs connect devices; WANs connect entire networks.
 - ! ◦ LAN and WAN differ in Layer 1 and Layer 2 methods, topologies, and media.
 - LAN wiring is private; WAN wiring is usually provided by public service providers like AT&T or Comcast.

WAN Essentials – (conts)

- **WAN site:** Individual geographic locations connected by WAN .
- **WAN link:** Connection between one WAN site and another site .
- ! • **Data Terminal Equipment (DTE):**
 - It is the customer's device at the end of the WAN.
 - The customer's device, works on the LAN.
- ! • **Data Communications Equipment (DCE):**
 - Service provider's device, works on the WAN side and handles the connection.
 - It communicates with the WAN.

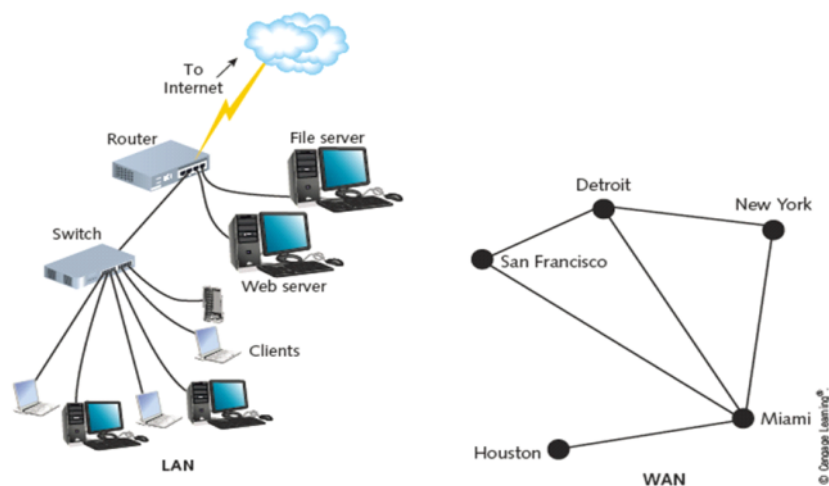


Figure 11-1 Differences in LAN and WAN connectivity

LAN (Local Area Network) !

- Shows a **single physical location** like an office.
- Devices are connected to a **switch**.
- The switch connects to a **router**, Router connects LAN to the internet.
- All connections are within the same building or local area.

WAN (Wide Area Network) !

- Shows multiple **cities** (San Francisco, Detroit, New York, etc.).
- Each city represents a **WAN site** (a local network at each location).
- These sites are connected using **WAN links**, forming a **wide area network**.
- Sites are linked to share data remotely.

Summary:

- **LAN** = local, short-distance connections (internal office network)
- **WAN** = connects LANs over long distances (between cities or countries)



WAN TOPOLOGIES



WAN topologies are different from LAN topologies

- They cover longer distances.
- They serve more users.
- They handle heavier network traffic.

Bus Topology

- Each site connects in a **line**, to a maximum of **two other sites**.
- Sites depend on each other to send and receive data.
- **Best for small WANs** with dedicated connections.
- **Not scalable** (hard to expand).

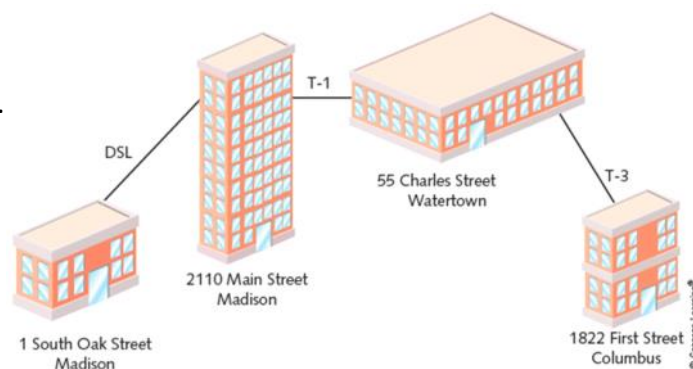


Figure 11-3 A bus topology WAN

Ring topology

- Each site connects to two others in a ring.
- Connects whole locations, not just local devices.
- Uses backup rings to reroute if a site fails.
- Expansion is hard and costly.
- **Best use ;**
- Connecting maximum five locations.

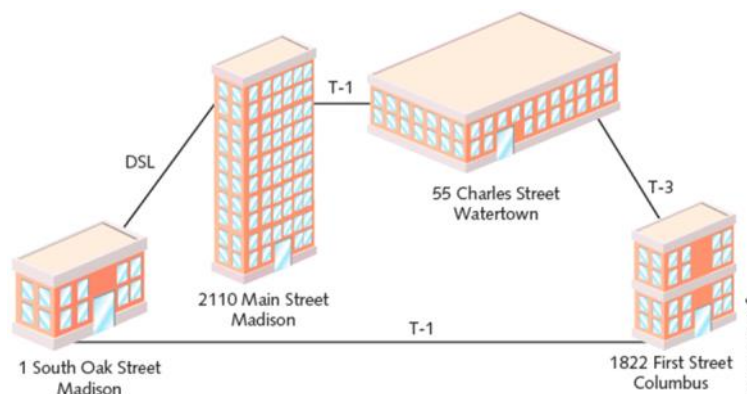


Figure 11-4 A ring topology WAN

Star Topology

- **Star WAN**
 - One site works as the **main hub** for all others
 - Each pair of sites uses a separate route for data
- **Advantages**
 - A single broken link affects only one site
 - Data travels faster between two sites
 - Easy and cheap to expand
- **Drawback**
 - If the central site fails, the whole WAN goes down

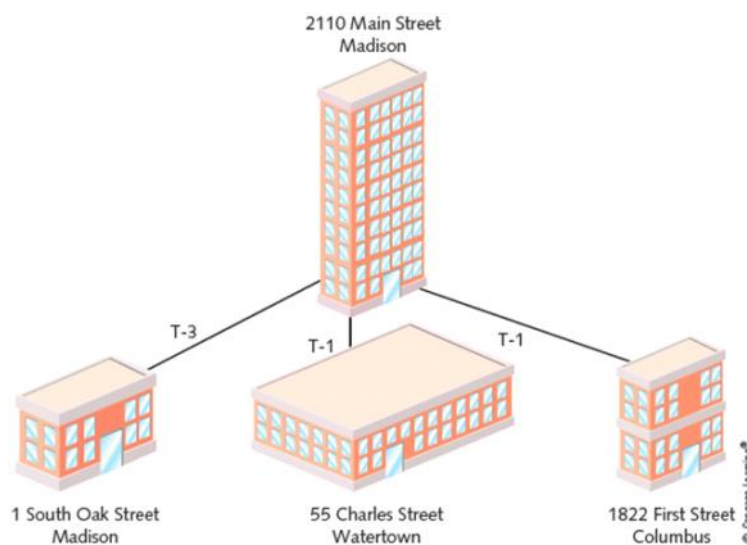


Figure 11-5 A star topology WAN

Mesh Topology

- **Mesh WAN**
 - Many sites are directly connected.
 - Data goes straight from source to destination
 - Routers can redirect data easily, quickly.
 - Most fault-tolerant WAN type.

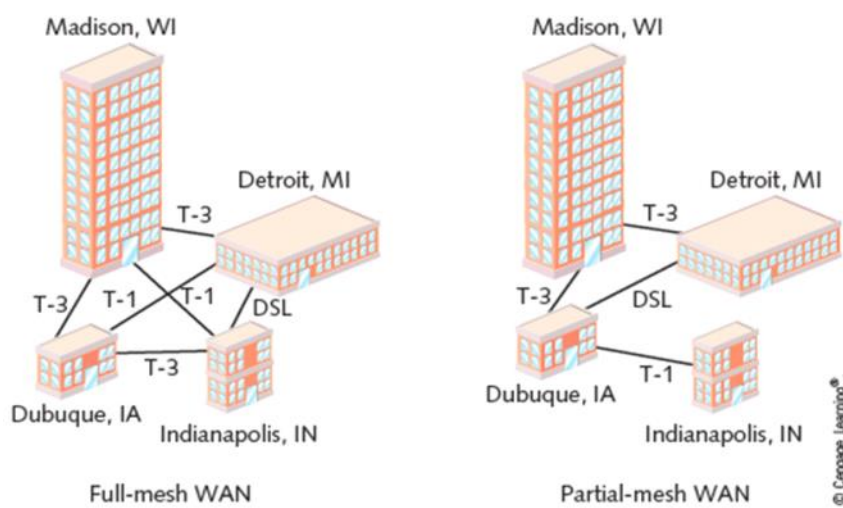


Figure 11-6 Full-mesh and partial-mesh WANs

- **Types of Mesh**
 - **Full-mesh:** All sites connect to each other (very reliable, but expensive)
 - **Partial-mesh:** Some sites connect directly (cheaper, less reliable)

Tiered topology WAN

- Sites are connected in star or ring shapes at different levels.
- Connection points are grouped in layers like a hierarchy.
- **Flexibility**
 - Allows different designs and is practical.
 - Must consider location, usage habits, and future growth.

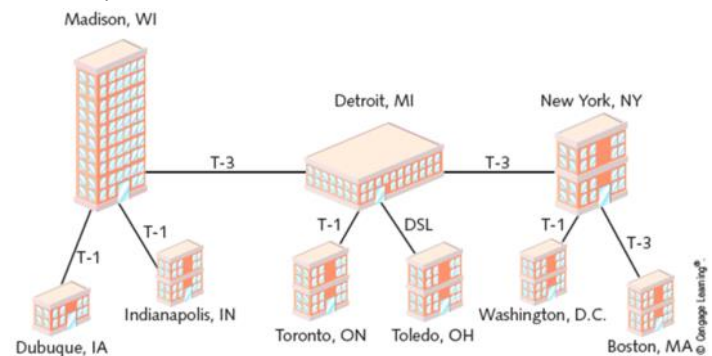


Figure 11-7 A tiered topology WAN

! • PSTN (Public Switched Telephone Network)

- It's a global phone system made of wires and telecom equipment.
- Also called POTS (Plain Old Telephone Service).
- Originally: analog traffic
- Today: digital data, controlled by computer

• CO (Central Office)

- Place where phone companies end their lines.
- It connects and transfers calls to other locations.

PSTN Components

- **Local Loop:**
Cable that connects homes or businesses to the nearest CO.
- **NIU (Network Interface Unit):**
The point where the customer's connection ends.
- **FTTP (Fiber to the Premises):**
Uses fiber cables to link a building; it's type of digital local loop.
- **PON (Passive Optical Network):**
Uses fiber-optic lines to serve multiple users from one provider.

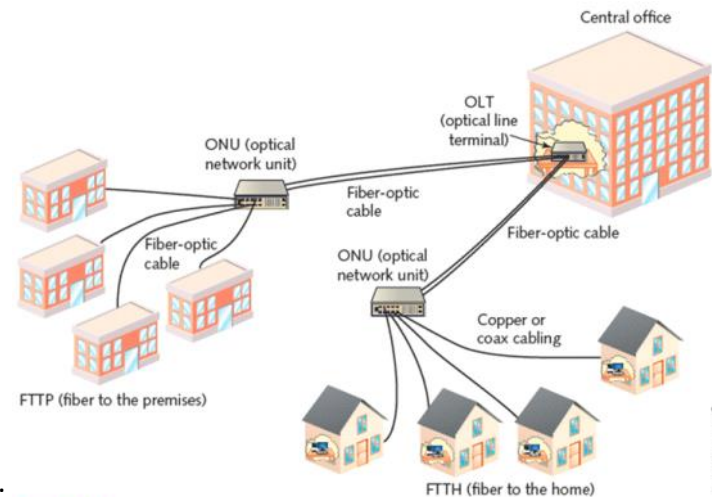


Figure 11-10 Passive optical network (PON)

OLT and ONU

- **OLT (Optical Line Terminal)**
 - Located at the service provider's main office.
 - Connects to many optical lines in a PON system.
- **ONU (Optical Network Unit)**
 - Sends signals to end users using fiber cables.
 - Can also use copper or coaxial cables.



Frame Relay

- **Frame Relay:**
 - It is a group of **Layer 2 (Data Link layer)** protocols.
 - Layer 2 protocols first made for fast packet-switching over ISDN.
 - Now used as a Data Link protocol for many types of media,circuit.
- **DLCI (Data-link Connection Identifier):**
 - Helps routers choose the correct circuit for data.
- **Connection Type:**Frame relay is a **connection-oriented** protocol.



◆ Two types of virtual circuits in Frame Relay:

- **SVC (Switched Virtual Circuit):** Connection established when needed and ends after data transmit is done.
- **PVC (Permanent Virtual Circuit):** Connection is established before,Connection is always ready, even before sending data. Connection keeps going on after the transmit.



◆ Advantages:

- You pay only for the bandwidth you use.
- Cheaper than most other WAN technologies.

DSL (Digital Subscriber Line)

- Works over the PSTN network
- Competes with T-1 and broadband cable
- Needs repeaters for long-distance use
 - Best for WAN local loop
- Supports multiple data and voice channels
 - All over one line
- Uses advanced modulation methods
 - Data changes the signal's strength or phase
 - Amplitude or phase modulation

• Asymmetrical DSL

- One direction has more bandwidth.
- Downstream speed is higher than upload.
- Best for: video calls, web browsing.

• Symmetrical DSL

- Equal upstream and downstream speeds.
- Types: HDSL, SDSL, SHDSL.
- Best for: heavy uploading/downloading.



Types of DSL

- xDSL means all types of DSL
 - ADSL, G.Lite, HDSL, SDSL, VDSL, SHDSL
- Symmetrical vs. Asymmetrical DSL
 - **Downstream:** Data goes from provider to customer
 - **Upstream:** Data goes from customer to provider

Modulation

- **Some DSL types** create narrow channels in high frequencies
 - Need splitters at both ends to separate voice and data
- **Others** need filters to block high-frequency signals from reaching phones.
- **Some DSL types** use different wire pairs than voice signals
 - Use extra wires already inside the phone cable

DSL Capacity and Line Length

- DSL types vary by speed and how far signals can travel.
- **VDSL** can reach up to 52 Mbps but only works well up to 1000 feet.
- **ADSL** is the most commonly used DSL type.
 - Latest version: **ADSL2+M**
 - Offers up to **24 Mbps download** and **3.3 Mbps upload** speed.



ATM (Asynchronous Transfer Mode)

- **Works at the Data Link layer**
- **Uses asynchronous communication**
 - Devices don't follow fixed timing rules
 - Timing must be specified for each transmission
 - Each character is sent with start and stop bits
- **Defines framing for the Data Link layer**
- **Fixed packet size**
 - Each cell = 48 data bytes + 5-byte header
- **Small packets increase overhead**
 - May reduce overall throughput
 - High cell efficiency helps balance the loss
- **Uses virtual circuits**
 - Considered a packet-switching method
 - Combines benefits of circuit switching
 - Ensures reliable data transmission
- **Supports guaranteed QoS (Quality of Service)**
 - Useful for applications needing consistent timing (e.g. voice, video)

ATM and frame relay are wan protocols.

- **ATM (Asynchronous Transfer Mode):**
 - Uses fixed-size cells for data transfer.
 - Provides reliable service, especially for voice and video.
 - Supports virtual circuits.
 - Guarantees quality of service (QoS).
- **Frame Relay:**
 - Uses variable-size frames for data transfer.
 - Simpler than ATM.
 - Still provides efficient data transfer across WANs.
 - Supports some level of QoS.

Summary

- Different businesses may require different WAN types.
- **Bus topology:** Each site connects to a maximum of two others in a line.
- **Ring topology:** Each site connects to two others, forming a circular path.
- **Star topology:** One central site connects and manages communication between several others.
- **Mesh topology:** Many sites are directly connected to each other for high redundancy.